Project 1

Duration : 3 weeks

Q1. Write a code that uses the vortex-panel method to compute the lift coefficient, leading-edge pitching-moment coefficient, and quarter-chord pitching-moment coefficient at a specified angle of attack on any NACA 4-digit airfoil. Also the code should be able to plot pressure coefficients on the upper and lower surface of airfoil and streamlines. For debugging purposes, the A matrix, B vector, and Gamma vector solutions for a NACA 2412 airfoil (with 10 points and an open trailing edge) at zero angle of attack can be viewed here. Results for this example should give the following:

$$\tilde{C}_L = 0.23983788991589367$$

$$\tilde{C}_{m_{LE}} = -0.11189872330227915$$

$$\tilde{C}_{m_{c/4}} = -0.051939250823305735$$

Compare all your results with XFOIL, where this is an introductory tutorial.